MScFE 610: Financial Econometrics

1 Introduction

Outliers—observations that deviate substantially from the majority—can distort parameter estimates, leading to biased inference and unreliable predictions [1]. OLS is especially sensitive since it minimizes squared residuals, giving disproportionate weight to extreme observations [2].

2 Sensitivity of Regression Models to Outliers

OLS solves

$$\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2,$$

so observations with large residuals or leverage exert strong influence [3].

Vertical outliers and high-leverage points both distort estimates, making OLS non-robust [4].

Influences:

- **Slope estimates**: extreme points can pull the regression line toward themselves.
- **Intercept estimates**: outliers may shift the regression line upward or downward.
- **Standard errors**: can be inflated or deflated, leading to unreliable inference.
- **Goodness-of-fit measures**: metrics such as R^2 may become misleading.

3 Illustration

For $y = 2 + 3x$, clean data yields $\hat{\beta}_1 \approx 3$. Adding $(10, 1000)$ produces $\hat{\beta}_1 \approx 30$, showing outlier influence.

4 Handling Outliers

4.1 Diagnostics

- **Standardized-residuals**(values exceeding ± 3 are typically flagged).
- **Cook's-Distance**(quantifies the influence of an observation) [3].
- **Leverage-statistics** (identifies high-leverage points).

4.2 Solutions

- **Winsorization** (capping extreme values).
- **Robust regression** (Huber loss, LAD regression) [1].
- **Transformations** (e.g., logarithmic scaling).
- **Outlier removal** (when justified and supported by domain knowledge).

5 Simulation Framework

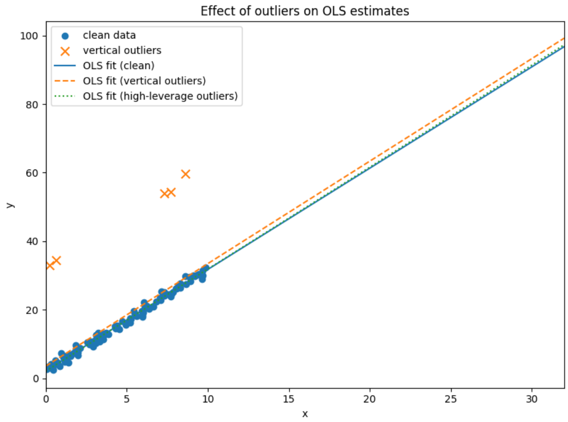

Figure 1: Effect of outliers on OLS regression estimates.

Estimated coefficients:
True coefficients: intercept = 2.000, **slope** = 3.000
OLS(clean data): intercept = 2.215, **slope** = 2.954
OLS(vertical outliers): intercept = 3.547, **slope** = 2.990
OLS(high-leverage outliers): intercept = 2.103, **slope** = 2.975

6 Stationarity and Unit Roots

A stationary series has constant mean/variance and lag-dependent autocovariances [5]. Non-stationarity leads to spurious regressions [6].

AR(1):

$Y_t = \rho Y_{t-1} + \varepsilon_t$

- $|\rho| < 1$: stationary
- $\rho = 1$: random walk
- $|\rho| > 1$: explosive

Tests:

- **Augmented Dickey-Fuller (ADF) test:** null hypothesis of a unit root [8].
- **Phillips-Perron test:** robust to heteroskedasticity and autocorrelation [9].
- **KPSS test:** null hypothesis of stationarity [10].

6.1 Addressing Unit Roots

If a series exhibits a unit root, possible remedies include:

- **Differencing:** $\Delta Y_t = Y_t - Y_{t-1}$.
- **Transformations:** logarithms or growth rates.
- **Cointegration methods:** Engle–Granger or Johansen tests for series with shared stochastic trends [11].

7 Why $\rho = 1$ Matters

Explosive ($\rho > 1$) processes are unrealistic and easily spotted. Random walks ($\rho = 1$) mimic trending data, causing spurious regression [12].

8 Simulation Illustration

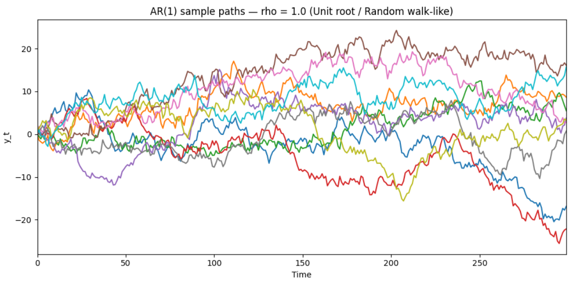

Figure 2: AR(1) with $\rho = 1.0$ (random walk).

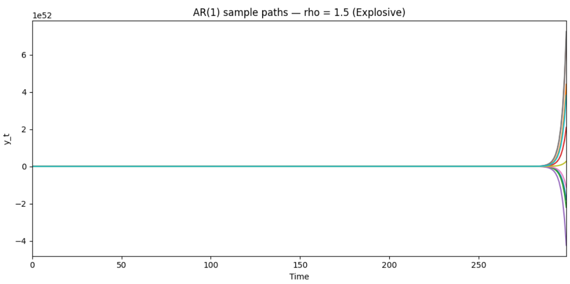

Figure 3: AR(1) with $\rho = 1.5$ (explosive).

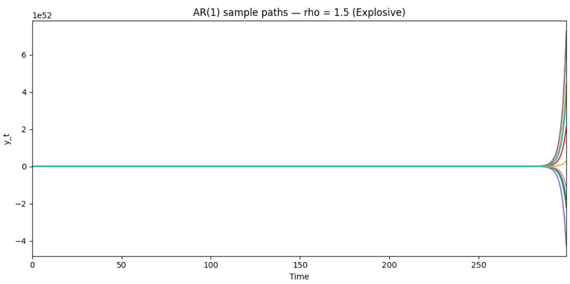

Figure 4: Comparison: $\rho = 0.8$ stationary, $\rho = 1$ unit root, $\rho = 1.5$ explosive.

References

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